

Sujan Thapa

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Summary

Versatile Software Developer specializing in the design and delivery of scalable SaaS solutions. Expert at architectural modernization, I recently worked for the transition of independent client applications into a centralized multi-tenant platform. By combining deep full-stack proficiency with an active focus on **Machine Learning and Artificial Intelligence**, I develop innovative software that solves real-world technical challenges with precision and reliability.

Professional Skills

Languages: JavaScript, TypeScript, Python, SQL, C++, PHP

Databases & ORMs: MySQL, PostgreSQL, MongoDB, MSSQL, Prisma, Sequelize

Frameworks & Libraries: ReactJS, NodeJS, ExpressJS, Fast API, Flask

Tools & Technologies: Git, GitHub, REST APIs, Web Sockets, Tailwind CSS

Core Concepts: System Design, Microservices, MVC Architecture, Agile Methodology

Experience

Paper Bank Solutions Pvt. Ltd.

Junior Full Stack Developer

Kathmandu, Nepal

Dec 2025 – Present

Projects Worked On: Document Management System (DMS)

- Architected the transition of multiple fragmented project codebases into a unified **SaaS platform**, consolidating unique client features into a single, scalable multi-tenant architecture.
- Troubleshoot and resolve critical issues within the core DMS application to ensure high system availability and uptime.
- Develop and enhance full-stack features using **Node.js, Express, React, and SQL**, focusing on system reliability and performance.
- Collaborate with clients to gather technical requirements and implement tailored DMS modules that solve specific business workflows.
- Optimize frontend architecture using **Tailwind CSS** and **React** to improve user experience and interface responsiveness across the platform.

Event Management Application

- Engineer scalable **RESTful APIs** using **Express** with **PostgreSQL**, following modular architecture for maintainable backend systems.
- Optimize data modeling and query performance by utilizing both **TypeORM** and **Drizzle ORM** across diverse application use cases.
- Enhance system throughput by implementing **Redis caching**, database indexing, and performance tuning to significantly reduce API response times.
- Design and secure ecosystem features including **Expo-based mobile components** using **JWT** and **Role-Based Access Control (RBAC)**.

eSatya

Blockchain Fellow

Kathmandu, Nepal

May 2025 – Aug 2025

- Studied blockchain fundamentals, smart contracts, and decentralized applications (dApps).
- Developed smart contracts for the **GitHub Bounty Dispenser** project to automate rewards.
- Learned about blockchain security, real-world use cases, and the mechanism of **Rahat** software.

ICT Matrics Pvt. Ltd.

PHP Developer

Kathmandu, Nepal

Jan 2025 – May 2025

- Developed and maintained core features for **Saral Khata** accounting software using **PHP, MySQL, and JavaScript**, focusing on financial data accuracy and user-centric design.
- Engineered **AJAX-based** asynchronous interactions to reduce page load times and implemented secure **session-based authentication** to protect sensitive user financial records.
- Optimized complex **SQL relational structures** and indexing strategies to ensure consistent and reliable management of high-volume accounting transactions.
- Collaborated with senior developers to integrate reporting modules, allowing users to generate automated balance sheets and tax summaries with precision.
- Streamlined data entry workflows by implementing real-time input validation and error-handling mechanisms, reducing data entry errors by end-users.

Premium Technologies Pvt. Ltd.

Kathmandu, Nepal

Customer Support Representative Intern

Feb 2022 – Apr 2022

- Provided technical support and resolved customer queries regarding **RDLC-based cheque printing systems**, ensuring seamless financial operations for clients.
- Diagnosed and troubleshooted software configuration issues, gaining early exposure to technical debugging and system-client interactions.
- Assisted in testing software updates and documented recurring bugs, contributing to product stability and user satisfaction.

Education

CITE College, Purbanchal University

Kathmandu, Nepal

Bachelor of Information Technology

CGPA: 3.71

- Graduated among the **Top 5%** of the batch, with a strong focus on core computer science fundamentals and software engineering principles.
- Developed expertise in programming, software development, and the practical application of **Machine Learning and AI models** within a professional IT context.

eSatya

Kathmandu, Nepal

Blockchain Fellowship

Graduated 2025

- Studied blockchain fundamentals, smart contracts, and decentralized applications.

Projects

Sajilo Aawas – Rental Management

Python, Fast API, JS, KNN

- Developed an end-to-end online rental platform designed to bridge the gap between tenants and property owners, streamlining the property discovery and booking process.
- Integrated the **K-Nearest Neighbors (KNN)** algorithm to provide intelligent, location-based room recommendations, ensuring users find relevant housing options in close proximity.
- Optimized data retrieval and system performance by implementing a dictionary lookup method for efficient handling of large property datasets and user search queries.

Fashion Product Recommender System

Python, Django, ResNet-50

- Engineered a visual search engine using **Python and Django** that allows users to upload images and receive instant suggestions for similar fashion products.
- Leveraged the **ResNet-50** architecture and Convolutional Neural Networks (CNN) to perform deep feature extraction, capturing intricate visual patterns for high-accuracy matching.
- Implemented a similarity-matching logic using KNN to compare extracted image embeddings against a product database, providing a seamless and automated shopping experience.

GitHub Bounty Dispenser (dApp)

Solidity, Ethereum, JavaScript, GitHub API

- Engineered a decentralized application (dApp) that automates reward distributions for open-source contributors by linking **GitHub pull request merges** to **Ethereum smart contract** payments.
- Developed and deployed **Solidity** smart contracts to securely escrow bounty funds, ensuring transparent and trustless payouts via automated trigger logic.
- Bridged Web2 and Web3 workflows by integrating the **GitHub API** to verify contributions, eliminating manual administrative overhead and ensuring immutable record-keeping on the blockchain.
- Implemented a **monorepo** architecture to efficiently manage the blockchain backend, integration logic, and frontend components within a unified development environment.

Smart Car (Embedded Project)

C++, Arduino, Sensors

- Developed an autonomous car system using **C++** and ultrasonic sensors to detect and navigate around physical obstacles in real-time.
- Programmed automated steering logic to ensure the vehicle maintains safe navigation without human intervention through continuous sensor feedback loops.

Additional

Languages: English, Nepali

Interests: Camping, Video Games, Sports, Teaching